INCH-POUND
MIL-DTL-32687
21 September 2021

DETAIL SPECIFICATION

WASHERS, CARBON OR ALLOY STEEL, ROUND, ZINC-NICKEL PLATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification covers custom carbon or alloy steel round washers (see 6.1) that are not covered by industry standards (see 6.4). This detail specification covers the application of Zinc-Nickel plating in lieu of either cadmium or hexavalent-chromium. The individual specification sheets detail cross references to prior cadmium or hexavalent chromium containing washers.

2 APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

MIL-PRF-32647	-	Zinc-Nickel Electroplating for Fasteners
MIL-DTL-32687/1	-	Washer, Flat, Alloy Steel, Round, Hardened, Zinc-Nickel Plated
MIL-DTL-32687/2	-	Washer, Flat, Alloy or Carbon Steel, Round, Hardened, Zinc-Nickel Plated, Extra Large

Comments, recommendations, additions, deletions, clarifications etc. and any data that may improve this document should be sent to U.S. Army Ground Vehicle System Center, ATTN: FCDD-GVS-SAT, MS #268, 6501 E. 11 Mile Road, Detroit Arsenal, MI 48397-5000 or emailed to usarmy.detroit.devcom-gvsc.mbx.standardization@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.dla.mil/.

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

ASME Y14.5	-	Dimensioning and Tolerancing
ASME-B18.18	-	Quality Assurance for Fasteners

(Copies of this document are available online at https://www.asme.org.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E384	-	Standard Test Method for Knoop and Vickers Hardness of Materials
ASTM F1470	-	Standard Practice for Fastener Sampling for Specified Mechanical Properties and Performance Inspection

(Copies of this document are available online at https://www.astm.org.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE AMS2671	-	Copper Brazing, Corrosion and Heat Resistant Steels and Alloys.

(Copies of this document are available online at https://www.sae.org/publications/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern (see 6.2).

3.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Edges and corners.

3.3.1 Edges. Unless otherwise specified, edges shall be broken R .02 +.02/-.00.

3.3.2 Corners. All edges and corners shall be free from burrs.

3.4 Finish. Finish shall be Zinc-Nickel plated in accordance with MIL-PRF-32647, Type K20.

3.4.1 Conformance. Certificates of compliance to paragraph 3.4 shall be provided to verify conformance.

3.5 Dimensions and tolerances. Dimensional limits shall apply after plating. Dimensioning and tolerances shall be in accordance with ASME Y14.5.

3.6 Workmanship. Washers shall be free from surface contamination, tool marks and other imperfections which may adversely affect serviceability.

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4 VERIFICATION

4.1 Inspection. Quality Assurance Provisions per Category 3 of ASME B18.18 are required.

4.1 Inspection of packaging. The sampling and inspection of the preservation, packing and container marking shall be in accordance with requirements of 5.1.

4.2 Finish and hardness inspection.

4.2.1 Protective coating and surface treatment inspection. Samples shall be inspected for adequacy of plating in accordance with MIL-PRF-32647.

4.2.2 Hardness inspection. Samples taken as specified in ASME B18.18 Category 3 sample size "C" shall be tested for surface hardness in accordance with ASTM E18.

5 PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in- house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6 NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Intended use of flat metal, round, washers are as follows:

a. Even distribution of the fastener's load.
b. Reduce friction of the fastener during the torqueing process.
c. Protect components from damage during the torqueing process.
d. Used as spacers.
e. May also be used in electrical terminal applications.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification and the applicable specification sheet.
b. Applicable specification sheet part number (see 3.1).
c. Protective finish required (see 3.4).
d. Packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

Corrosion
Decarburization
Descaling

6.4 Difference from industry standards. At the time of this standards publication, the specified washers differ dimensionally from similar industry standard washers as found in ASME B18.24, "Part Identifying Number (PIN) Code System Standard for B18 Fastener Products". The user is encouraged to use ASME standard washers when applicable.

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Army – AT
Navy - AS
Air Force – 70

Review Activity:
Army - AV, CR4, MI
Navy - SH, MC, YD
Air Force – 20

Preparing Activity:
Army - AT

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